Event ends in 1 day 4 hours 8 minutes.

Prerequisites 📋

Important
- Use Kiro CLI for this section
- Complete Setting Up Kiro

Situation 🔍

As a cloud engineer, you've been tasked by the management team to help with cost optimization. You noticed in Cost Explorer that there was a high volume of data going through the NAT gateway, and you need to investigate this issue. We have default VPC flow logs set up 📊

1. Use Kiro to Query Default VPC Flow Logs 📊

1. In your VS Code Server environment, create a new folder to separate content from other files inside VS Code Server:

```
cd /Workshop
mkdir default-flowlog-analysis
cd default-flowlog-analysis
```

2. Open Kiro Chat:

```
kiro-cli chat
```

3. Let's ask Kiro CLI to identify the top 50 IP traffic sources going through the NAT gateway. (Optional) You can also use this prompt in your environment by changing the VPC ID and AWS Region. Ask Kiro CLI using the following prompt 🔎:

```
What is the top 50 traffic going through NAT Gateway ENI in the last 1 hour
VPC name: samplevpc-VPC
Region: us-east-1
Query traffic from VPC flow log
Make sure to check log format to see the field
Output to traffic_table.csv
```

traffic_table.csv

/Workshop/default-flowlog-analysis/traffic_table.csv

```
Source IP,Destination IP,Protocol,Bytes,Packets,Connections
10.0.1.103,10.0.3.136,TCP,15572041,7907,88
10.0.3.136,10.0.1.103,TCP,15494788,6315,85
10.0.4.124,10.0.1.103,TCP,14470248,6712,82
10.0.1.103,10.0.4.124,TCP,14464469,7376,89
....
```

4. Let's make sense of these IPs using Q. AWS provides IP ranges of AWS services, which may change over time. Customers can find the latest list of IP ranges from this JSON document: https://ip-ranges.amazonaws.com/ip-ranges.json 🔗

```
Download the full list of AWS IP ranges from https://ip-ranges.amazonaws.com/ip-ranges.json
then create a filtered list of AWS IP addresses.
* Exclude "service": "AMAZON"
* Filter IP for region: us-east-1
Output to filtered_aws_ips.csv
```

filtered_aws_ips.csv

default-flowlog-analysis > filtered_aws_ips.csv

```
99.77.240.0/24,CHIME_VOICECONNECTOR
54.251.31.128/26,ROUTE53_HEALTHCHECKS
54.255.254.192/26,ROUTE53_HEALTHCHECKS
16.12.48.0/21,S3
52.219.32.0/21,S3
....
```

5. Let's ask Kiro CLI to tag IPs in the table using data from AWS-provided IP ranges 📌:

```
Tag IPs in traffic_table.csv using filtered_aws_ips.csv to understand what each IP is associated with
* VPC IP
* AWS Service Name
* Internet
Output as CSV traffic_tagged.csv
```

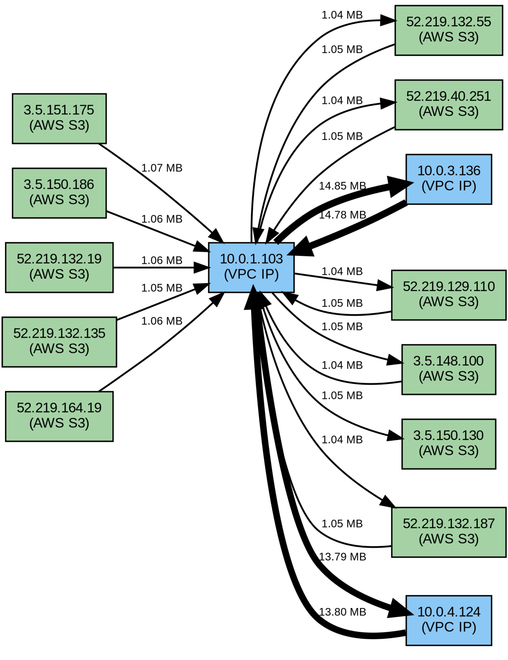
traffic_tagged.csv

default-flowlog-analysis > traffic_tagged.csv

```
Source IP,Destination IP,Source Type,Destination Type,Protocol,Bytes,Packets,Connections
10.0.1.103,10.0.3.136,VPC IP,VPC IP,TCP,15572041,7907,88
10.0.3.136,10.0.1.103,VPC IP,VPC IP,TCP,15494788,6315,85
10.0.4.124,10.0.1.103,VPC IP,VPC IP,TCP,14470248,6712,82
10.0.1.103,10.0.4.124,VPC IP,VPC IP,TCP,14464469,7376,89
...
```

6. Visualize the data to see the sources and destinations going into and from the NAT gateway 📊:

```
Using traffic_tagged.csv, create a network visualization diagram.
Node is IP address, with label IP, Tag.
Edge is a directed graph between src address and dst address
Edge weight based on total data transfer volume, edge thickness proportional to data transfer volume. Use graphviz library to visualize.
```



7. Let's ask Kiro CLI to group the nodes together by category to better understand what's going on 🗃️:

```
Can you group IPs with the same tag closer together
```

8. **Conclusion** 💡 - Using visualization, you can see that the majority (top 20) of network traffic through the NAT Gateway is from EC2 and S3. You can use VPC Gateway endpoints for Amazon S3 🔗 to access Amazon S3 from your VPC using gateway VPC endpoints, bypassing the NAT Gateway. After you create the gateway endpoint, you can add it as a target in your route table for traffic destined from your VPC to Amazon S3.

🔥 There is no additional charge for using gateway endpoints.

2. Use Kiro to Create VPC Endpoint for S3 🗡️

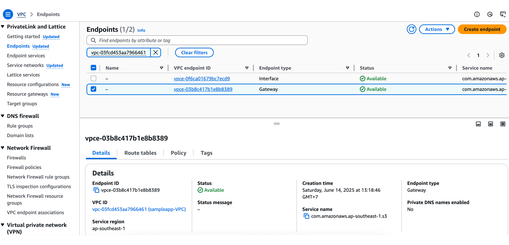
1. Let's start a new Kiro session. To terminate the existing session, use `/quit` 🚪:

```
kiro-cli chat
```

2. Let's ask Kiro CLI to create a VPC endpoint 🔗:

```
Set up S3 and DynamoDB VPC endpoints
VPC name: samplevpc-VPC
Region: us-east-1
```

3. Check in the console to see if the S3 VPC endpoint is properly configured ✅:



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